

Hillsborough River

Shoreline length: 146 km



Shore Types

Hillsborough River is a tidal estuary with a majority of wetland and low plain shores.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
15	2	0	1	82	0	0	0



City View Drive, south shore of Hillsborough River



Mount Stewart bridge(left), Confederation trail (middle) and wastewater treatment plant lagoons (right)

Governance, Infrastructure, and Economy

Hillsborough River SMU falls within several jurisdictions: the Town of Stratford (~1%), with the vast majority of the SMU including Mount Stewart, the Abegweit First Nation's Scotchfort Indian Reserve No. 4 (managed under First Nations authority), and an unincorporated area (~96% combined).

Areas with road infrastructure includes Miles, Clarks, and Fullertons Creeks; the Mount Stewart and Confederation Trail bridges over the Hillsborough River; and the Fort Augustus and Donagh Road crossings at the Augustus River.

Key infrastructure in this SMU includes the Abegweit Biodiversity Enhancement Hatchery. Other infrastructure includes Route 260 (Frenchfort Road at Miles Creek); Route 218 (Portage Road, east of McNally Shore Road); Main Street (Mount Stewart Bridge of Hillsborough River); the intersection of Routes 352 and 3TCH at Stratford/Cherry Hill; Route 21 (Fort Augustus Road and bridge at Clarks Creek); the Fort Augustus Road crossing at Augustus River; Route 257 (Donagh Road crossing at Augustus River); Fort Augustus Road at Mermaid; and Route 21 (Bunbury Road and bridge at Fullerton's Creek), and wastewater treatment ponds in Mount Stewart.

No major coastal economic drivers are noted. Legacy land use constraints include some densely developed small lots, and some unpaved roads.

Natural Assets and Heritage

Hillsborough River is home to the Glenfinnan Wildlife Management Area, Mount Stewart Wildlife Management Area, and Pisquid Wildlife Management Area. Protected and recreational sites include the Acadian Forest Park, Duffy's Beach, and the Fort Augustus Lions Club ball and soccer fields.

Planning and Policy Framework

Provincial rules apply in the unincorporated portion of this SMU and in Mount Stewart (a municipality with no planning). Land use and development standards include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in

regulations, but coastal hazard assessments are typically conducted. Stratford has municipal planning authority. Current planning documents apply watercourse and wetland regulations in all zones, including an O2 Reserve Zone — not shown on the zoning map — whose boundaries are interpreted to include any area defined as a Wetland or Watercourse, plus the area within 60 metres (197 feet) of a Wetland or Watercourse. Draft documents currently out for consultation would add vertical elevation requirements for habitable floors. The Abegweit First Nation manages land use within the Scotchfort Reserve under its own authority, separate from municipal or provincial planning.

Coastal Flood and Erosion Risk

Coastal flooding is the prevailing hazard for the area. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential shoreline change or salt marsh migration to future time horizons.

Number of at-risk buildings [N. per km shoreline]

	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
SMU Hillsborough River				
Flood risk	50 [<1/km]	69 [<1/km]	90 [1/km]	147 [1/km]
Erosion risk	NA	22 [<1/km]	27 [<1/km]	97 [1/km]
PU Mount Stewart				
Flood risk	22 [15/km]	29 [19/km]	32 [21/km]	43 [29/km]
Erosion risk	NA	4 [3/km]	4 [3/km]	23 [15/km]

Sediment transport - Sediment transport rates and associated downdrift risk across the SMU are low.

Proposed SMP Strategies

The primary strategy for Hillsborough River is No Active Intervention until flood risk warrants managed realignment options.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	NAI	MR	MR
Rationale	Mid- to long-term MR will reduce risk to the limited number of vulnerable assets while preserving natural shorelines, including wetland habitat with increasing flood hazard.		
Proposed PU strategy			
Mount Stewart	MR	MR	MR
Rationale	MR will reduce risk to the limited number of vulnerable assets, allowing salt marsh migration up the estuary. In the short term, floodproofing measures for vulnerable assets are recommended. Note: Wastewater treatment lagoons will need to consider flood mitigation measures		



Policy Unit:
Mount Stewart

LEGEND

- Policy Unit
- Shoreline Management Unit
- Net Longshore Sediment Transport
- High Hazard Area
- Flood Extent - Present Day (100-year Return)
- Flood Extent - 1.2m SLR (100-year Return)
- Property Boundary
- Building > 90ft²

0 450 900 1,800
Metres

Coordinate System:
NAD 83 CSRS PEI Double Stereographic

NOTES:

**COASTAL EROSION
AND FLOOD HAZARD**

Shoreline Management Unit:
Hillsborough River

Prince Edward Island
Île-du-Prince-Édouard